

Context Document: SPI-SPI Feature Space for Comparing Multivariate Time Series of Arbitrary Dimension

Purpose of This Document

This document provides complete methodological context for ongoing work toward an EEML 2026 extended abstract submission. It describes a method for embedding multivariate time series (MTS) of arbitrary spatial dimension M and temporal length T into a common feature space. This document covers: (1) the problem and motivation, (2) the baseline method (SPI-SPI space), (3) its known limitations, and (4) illustrative case studies. A companion document describes the proposed extension using geometric deep learning.

1. Problem Statement

1.1 Feature-Based Univariate Time Series Analysis

Feature-based time series analysis transforms a univariate time series $x \in \mathbb{R}^T$ into a fixed-dimensional feature vector $\mathbf{f} \in \mathbb{R}^K$ by computing K summary statistics (e.g., mean, autocorrelation, spectral entropy). This eliminates dependence on T , enabling comparison of time series of different lengths in a common K -dimensional feature space amenable to standard statistical and machine learning methods.

1.2 The Multivariate Extension Problem

For an M -channelled multivariate time series $\mathbf{X} \in \mathbb{R}^{M \times T}$, the spatial dimension M introduces a second axis of variation. Feature-based methods that summarise pairwise interactions between channels (producing, e.g., $M \times M$ matrices) eliminate dependence on T but remain sensitive to M . Specifically: if two MTS processes $\mathbf{X} \in \mathbb{R}^{M_1 \times T_1}$ and $\mathbf{Y} \in \mathbb{R}^{M_2 \times T_2}$ have $M_1 \neq M_2$, their pairwise interaction matrices live in different-dimensional spaces and cannot be directly compared.

Naïve solutions such as truncating to the minimum common channel count discard channels and, with them, potentially important dynamical regimes and inter-channel dependency structures. The problem is therefore: **how can we embed MTS of arbitrary M and T into a common feature space that preserves informative dependency structures?**

1.3 Existing Approaches

This problem is not entirely unaddressed. Several families of methods can compare graphs (or systems) of different sizes:

- **Graph kernels** (Weisfeiler-Leman subtree, random walk, graphlet kernels) define kernel functions between graphs of arbitrary size and are well-established in graph classification.
- **Global graph pooling in GNNs** (sum/mean/attention readout) produces fixed-dimensional embeddings regardless of node count.
- **NetSimile and similar feature-extraction approaches** compute distributions of local structural features (degree, clustering coefficient, etc.) and compare their statistics.

The method described here is a novel instantiation within this broader family, distinguished by: (a) using a diverse, interpretable library of pairwise interaction statistics as the input representation, and (b) exploiting the *relationships between* different statistical characterisations as the embedding, rather than structural graph features.

2. The SPI-SPI Feature Space (Baseline Method)

2.1 Pairwise Interaction Matrices (MPIs)

Let $\mathbf{X} \in \mathbb{R}^{M \times T}$ be an MTS with channels $\{X_t^{(i)}\}_{i=1}^M$. The Python library `pyspi` computes a library of $K > 250$ statistics of pairwise interaction (SPIs) between channels. For the k -th SPI, computing it on every ordered pair (i, j) yields an $M \times M$ adjacency matrix

$$\text{MPI}_k \in \mathbb{R}^{M \times M}, \quad [\text{MPI}_k]_{ij} = \text{SPI}_k(X^{(i)}, X^{(j)}).$$

We call this a **matrix of pairwise interaction (MPI)**. Geometrically, this is a weighted directed graph on M nodes, where node i represents channel $X^{(i)}$ and edge weight (i, j) is the value of the k -th SPI computed between channels i and j .

Computing all K SPIs yields a tensor $\mathcal{M} \in \mathbb{R}^{K \times M \times M}$ — a collection of K interaction graphs on the same M nodes.

Note on temporal invariance. Each SPI maps a pair of time series (of any length T) to a scalar. Therefore, $\mathcal{M}$ is invariant to T . Two MTS $\mathbf{X} \in \mathbb{R}^{M \times T_1}$ and $\mathbf{Y} \in \mathbb{R}^{M \times T_2}$ with $T_1 \neq T_2$ produce MPIs of identical dimension, enabling direct comparison of their interaction structures.

Remaining limitation. The tensor $\mathcal{M} \in \mathbb{R}^{K \times M \times M}$ still depends on M . Systems with different channel counts produce tensors of different size.

2.2 Constructing the SPI-SPI Feature Vector

To eliminate dependence on M , we compute a second-order summary: the correlation between pairs of SPIs across all pairwise channel interactions within a single MTS.

For the k -th MPI, extract the $\binom{M}{2}$ off-diagonal entries (or all $M(M-1)$ entries if the SPI is asymmetric) into a vector $\mathbf{v}_k \in \mathbb{R}^{M(M-1)/2}$. For two SPIs k and k' , compute

$$f_{kk'} = \text{corr}(\mathbf{v}_k, \mathbf{v}_{k'})$$

where `corr` is Pearson's product-moment correlation. This yields a scalar summarising how similarly the two SPIs rank the pairwise interactions within the system.

For K SPIs, we obtain a $\binom{K}{2}$ -dimensional feature vector

$$\mathbf{f} = \{f_{kk'}\}_{k < k'} \in \mathbb{R}^{\binom{K}{2}}$$

which we term the **SPI-SPI feature vector**, and the resulting space **SPI-SPI space**.

Key property. The dimension of $\mathbf{f}$ depends only on K (the number of SPIs chosen), not on M or T .

Therefore, any two MTS — regardless of their spatial or temporal dimensions — can be embedded into this common $\binom{K}{2}$ -dimensional space.

2.3 Interpretation

Each feature $f_{kk'}$ encodes the empirical relationship between two statistical characterisations of pairwise interaction across the channels of a single system. Comparing $\mathbf{f}$ across different MTS processes reveals similarities and differences in the **character** or **nature** of dependency structures, rather than in the strength of specific channel-pair interactions.

This is a critical distinction: SPI-SPI space is agnostic to *which* channels are coupled. It captures the statistical fingerprint of the coupling regime but discards the topological arrangement. This is both a strength (enabling comparison across different M) and a limitation (see §2.4).

2.4 Choice of Correlation Measure

The off-diagonal entries of an MPI are not statistically independent — entries sharing a node (e.g., $\text{SPI}_k(1, 2)$ and $\text{SPI}_k(1, 3)$) are confounded by the marginal properties of the shared channel. This ****network autocorrelation**** inflates the effective sample size when computing $\text{corr}(\mathbf{v}_k, \mathbf{v}_{k'})$, potentially leading to overconfident correlation estimates.

We use Pearson correlation rather than Spearman rank correlation for $f_{kk'}$. The rationale: rank transformation may corrupt subtle quantitative signatures — specifically, when the signal lies in the relative *magnitudes* of SPI values across channel pairs (not just their ordering), Spearman's rank space discards this information. Since the method's discriminative power relies on detecting fine-grained shifts in the covariation of SPIs, Pearson is preferred.

However, this choice should be validated empirically. A robustness check comparing Pearson, Spearman, and Kendall's τ for the inter-MPI correlation is warranted. Additionally, permutation-based significance testing or QAP (quadratic assignment procedure, standard in social network analysis) should be considered to account for the non-independence structure.

2.5 Known Limitations

1. ****Information bottleneck.**** Reducing an $M \times M$ adjacency matrix to a single scalar per SPI pair is aggressive. Graph topology — community structure, hub-spoke patterns, motifs, degree distribution — is entirely discarded. Two systems with identical coupling *character* but different network architectures (e.g., star vs. ring topology) are indistinguishable in SPI-SPI space.
2. **Scalability of feature space.** For K SPIs, the feature vector has $\binom{K}{2}$ entries. With $K = 250$, this is 31,125 features — high-dimensional relative to typical sample sizes, requiring careful regularisation or feature selection.
3. **No learned representation.** The features are hand-crafted. There is no mechanism to learn which SPIs or which inter-SPI relationships are most informative for a given task.

4. **Network autocorrelation in the correlation estimate.** As noted in §2.4, the non-independence of off-diagonal entries biases the correlation. This is a statistical subtlety that does not invalidate the method but requires careful treatment.
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3. Illustrative Case Studies

These case studies demonstrate the interpretive logic of SPI-SPI space on controlled synthetic systems. They serve as pedagogical motivation, not as the primary contribution.

3.1 Differentiating Linear, Nonlinear, and Non-Monotonic Dependencies

Setup. We construct an $M \times T$ MTS where each channel is a noisy, filtered copy of a latent autoregressive process

$$z_t = a \cdot z_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1),$$

passed through a nonlinear filter $g(z; \alpha)$ and corrupted by observation noise:

$$X_t^{(i)} = g(z_t; \alpha_i) + \eta_t^{(i)}, \quad \eta_t^{(i)} \sim \mathcal{N}(0, \sigma_\eta^2).$$

The filter $g(z; \alpha)$ is defined as: (1) rescale z_t to $[0, 1]$ via min-max normalisation, (2) map to the interval $[-\alpha, \alpha]$, (3) apply $\sin(\cdot)$, and (4) normalise to $[-1, 1]$ by dividing by the theoretical maximum $g_{\max} = \min(\sin(\alpha), 1)$. The normalisation preserves a constant signal-to-noise ratio across filter configurations.

Experimental logic. The parameter α controls the degree of nonlinearity:

- For $\alpha \ll 1$: $\sin(x) \approx x$ (linear regime).
- For α approaching $\pi/2$: the mapping becomes nonlinear but monotonic.
- For $\alpha > \pi/2$: the mapping becomes non-monotonic.

Each channel i draws $\alpha_i \sim \text{Uniform}(\pi/\epsilon, A)$ where ϵ is small (e.g., $\epsilon = 1/64$). The control parameter A governs the maximum nonlinearity across channels. As A increases toward π , a greater proportion of channels are filtered through nonlinear (and eventually non-monotonic) transformations.

Expected behaviour of $f_{kk'} = \text{corr}(r, \rho)$. In the linear regime, Pearson's r and Spearman's ρ agree closely across channel pairs, yielding $f_{kk'} \approx 1$. As more channels undergo nonlinear filtering, the Pearson MPI values degrade (violating linearity/Gaussianity assumptions) while Spearman values remain stable (rank-invariance under monotonic transformation). The result is a progressive decoupling: $f_{kk'}$ decreases as a function of A , with a sharper transition near $A = \pi/2$ where non-monotonicity begins to also degrade Spearman.

****Interpretation.**** The degree to which r and ρ are correlated across channel pairs (the feature $f_{kk'}$) provides a signature of the proportion and character of nonlinear interactions in the system. This illustrates how SPI-SPI features encode dependency *character* without reference to specific channel identities.

****Important caveat.**** The latent driver z_t is itself a non-monotonic autoregressive process. The method characterises the nature of *inter-channel* dependence, not the marginal dynamics of individual channels. The filter g modulates how each channel relates to the shared driver, and it is these inter-channel relationships that the SPIs (and hence SPI-SPI features) capture.

3.2 Temporal Misalignment via DTW and Euclidean Distance

****Setup.**** Channels are noisy, *temporally warped* copies of the latent driver z_t . The warping for channel i is generated by a random walk in $(t_{\text{orig}}, t_{\text{warp}})$ -space: at each timestep, with probability $p_{\text{step}}^{(i)}$, an "L-shaped" excursion deviates from the identity line. The excursion size is geometric: $s \sim \text{Geom}(q)$, with $q = 0.5$ (giving $\mathbb{E}[s] = 2$).

The warp intensity parameter is $p_{\text{step}}^{(i)} \sim \text{Uniform}[0, a]$, making a the single control parameter governing the degree of temporal misalignment across the MTS.

Warpedness measure. Per-channel warpedness is defined as the L_1 deviation from the identity alignment:

$$\mathcal{W}^{(i)} = \sum_{t=1}^T |t_{\text{orig},t} - t_{\text{warp},t}^{(i)}|.$$

The per-MTS aggregate $\mathcal{W} = M^{-1} \sum_i \mathcal{W}^{(i)}$ satisfies $\mathbb{E}[\mathcal{W}] \approx 2aT$, scaling linearly in a . The normalised form $\bar{\mathcal{W}} = (MT)^{-1} \sum_i \mathcal{W}^{(i)}$ satisfies $\mathbb{E}[\bar{\mathcal{W}}] \approx 2a$, depending only on a .

Expected behaviour of $f_{kk'} = \text{corr}(\text{ED}, \text{DTW})$. For unwarped signals ($a \approx 0$), DTW reduces to pointwise Euclidean distance (the optimal alignment path is the identity). As a increases, DTW's flexible alignment compensates for warping while Euclidean distance degrades. The decoupling of ED and DTW across channel pairs — captured by a decrease in $f_{kk'}$ — is a signature of the degree of temporal misalignment in the system.

3.3 Extension Beyond Toy Examples

In practice, individual features $f_{kk'}$ carry subtle signals, and it is the collective pattern across the full $\binom{K}{2}$ -dimensional vector that constitutes the system's dynamical signature. By selecting K SPIs spanning diverse statistical families — correlation measures, distance measures, information-theoretic quantities, spectral measures, causal/directed measures — the SPI-SPI feature space forms a "meshgrid" of statistical assumptions capable of jointly characterising complex dependency regimes.

Preliminary results. Clustering analyses in SPI-SPI space, visualised via PCA, UMAP, and t-SNE, show clear separation of synthetic MTS classes differing in dependency structure, with separation robust to variation in M and T . Formal benchmarks against baselines are pending.

4. Open Questions and Status

- The correlation choice for $f_{kk'}$ (§2.4) requires empirical validation.
- The method has not been benchmarked against graph kernel baselines.

- Real-world MTS experiments are needed.
- The known limitation of discarding graph topology (§2.5) motivates the geometric deep learning extension described in the companion document.